

Label-Preserving Signed Observables after Radial Completion: A Fixed-Shift Coprime-Shell Law, Uniform Short-Shift Masks, and the Type-II Gate

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July 2026

Abstract

Let $u(n) = \Lambda(n) - 1$, let $W \in C_c^1((0, \infty); \mathbb{R})$ and $G \in C_c^1(\mathbb{R}; \mathbb{R})$ be nonzero, and write every factor pair uniquely as $n = ak$, $m = bk$, $(a, b) = 1$. TPC-13 separated the joint labels $(\log(a/b), \log k)$ and proved diagonal-energy laws together with character-phase first moments. Here we evaluate one fixed-shift aggregate and close the short-shift first moment from character phases to the full bounded-mask ball.

The first result is at a prescribed fixed nonzero shift. The exact radial shell $k = 1$, with all primitive orientations retained, is

$$\mathcal{K}_h(X) = \sum_{\substack{a > 1, b \geq 1 \\ (a, b) = 1}} u(a)(\Lambda(ab + h) - 1)W(ab/X).$$

If

$$\beta_h = \prod_{p|h} \left(1 - \frac{p}{p^2 - p + 1}\right),$$

then the unitary-divisor Titchmarsh theorem of Biao Wang implies

$$\mathcal{K}_h(X) = \left(\frac{1}{\zeta(2)} - \beta_h\right) X I_0(W) \log X + O_{h,W}(X).$$

Thus a nontrivial signed GCD–Mellin radial shell is evaluated for every fixed h , including $h = 2$. It still sums all primitive directions b ; it does not isolate the endpoint $b = 1$ and gives no twin-prime conclusion.

The second result controls the full coefficient-space dual ball after a growing short-shift average. For $X^{2/15+\varepsilon} \leq H \leq X$ and every $A > 0$, the Guth–Maynard short-interval inputs give, provided all translated targets remain at height comparable with X ,

$$\left\| \frac{1}{H} \sum_q G(q/H) (\Lambda(\cdot + h_0 + q) - 1) \right\|_{\ell^2(r \asymp X)} \ll_{A, \varepsilon, G} X^{1/2} (\log X)^{-A}.$$

Consequently, uniformly for every $D \geq 1$, every bounded, shift-independent mask $|\omega(a, b, k)| \leq 1$ satisfies

$$\begin{aligned} & \frac{1}{H} \sum_q G(q/H) \sum_{\substack{a, b, k \geq 1, (a, b) = 1 \\ ak > D}} \omega(a, b, k) u(ak) \\ & \times (\Lambda(ak^2 + h_0 + q) - 1) W(ak^2/X) \ll_A X (\log X)^{-A}. \end{aligned}$$

uniformly in the mask. Taking $D = 1$, this includes the exact endpoint $b = k = 1$, individual radial layers, primitive rays, and arbitrary signed unions. It yields an averaged endpoint prime-pair asymptotic at the two-fifteenths scale, but no selected shift.

After recentering on the factorized target variable, the same theorem controls every bounded mask of Vaughan's hard packet under the short-shift average. Finally, an exact local-data minimax formula and a nearly sharp square-root divisor-signature dichotomy exhibit a prime/semiprime null direction. A balanced bilinear factor row detects that direction exactly; detecting it is an information test, not an estimate for the arithmetic packet. No fixed-shift prime-pair asymptotic, twin-prime lower bound, new level of distribution, or breach of sieve parity is claimed.

Keywords: GCD–Mellin labels; unitary divisors; Titchmarsh divisor problem; short-shift prime correlations; Vaughan identity; Type-II information; sieve parity.

MSC 2020: Primary 11N05, 11N37; Secondary 11N35, 11N32, 42A05.

1 Introduction and scope

The sequence represented here by TPC-7, TPC-12, and TPC-13 separated three questions that are often compressed into one. A complete multiplicative aggregate can be evaluated at a fixed shift, but may hide the factor scale. A Mellin coordinate can resolve primitive ratio rays, but a ratio alone collapses every radial multiple. Adding the GCD coordinate restores the full factor label, but exact inversion merely reveals the coefficient: it does not estimate the fixed-shift prime pair sitting at the endpoint. See Wang [10, 11, 13].

This paper takes the next step in two different quantifier regimes. First, we keep one exact radial shell and retain a prescribed fixed shift. Second, we keep an arbitrary signed label mask but average the shift over a growing window. The distinction is essential.

All asymptotics are as $X \rightarrow \infty$. The functions W, G and the fixed shift h in the coprime-shell theorem are held fixed; uniform parameters are identified explicitly.

Put

$$u(n) = \Lambda(n) - 1, \quad \eta(t) = \Lambda(t) - 1, \quad (1)$$

and fix nonzero real functions $W \in C_c^1((0, \infty))$ and $G \in C_c^1(\mathbb{R})$, with

$$\text{supp } W \subset [\alpha, \beta] \subset (0, \infty). \quad (2)$$

$$I_0(W) = \int_0^\infty W(v) dv, \quad I_0(G) = \int_{\mathbb{R}} G(v) dv. \quad (3)$$

Every factor pair has the unique representation

$$n = ak, \quad m = bk, \quad (a, b) = 1, \quad (4)$$

and the product center is

$$r = mn = abk^2. \quad (5)$$

1.1 The fixed-shift radial shell

The shell $k = 1$ consists of the coprime factor pairs (a, b) . At source cutoff $D = 1$, its signed residual is

$$\mathcal{K}_h(X) = \sum_{\substack{a > 1, \, b \geq 1 \\ (a, b) = 1}} u(a)\eta(ab + h)W(ab/X). \quad (6)$$

For a fixed nonzero integer h , define

$$\beta_h = \prod_{p|h} \left(1 - \frac{p}{p^2 - p + 1}\right), \quad (7)$$

where the product is over the prime divisors of $|h|$ and the empty product is 1. Our first main theorem is

$$\mathcal{K}_h(X) = \left(\frac{1}{\zeta(2)} - \beta_h \right) X I_0(W) \log X + O_{h,W}(X). \quad (8)$$

The proof is not a prime-pair estimate. Grouping by $r = ab$ turns the source coefficient into

$$\log \text{rad}(r) - 2^{\omega(r)} + 1. \quad (9)$$

The term $2^{\omega(r)}$ is the unitary divisor function. The fixed-shift prime target against this function is evaluated by the published unitary-divisor Titchmarsh theorem of Wang [9]. The remaining terms are elementary at the required precision.

For $h = 2$, (7) gives $\beta_2 = 1/3$ and the coefficient in (8) is

$$\frac{6}{\pi^2} - \frac{1}{3} > 0. \quad (10)$$

This is a fixed-2 theorem, but not a twin-prime theorem. The shell retains $k = 1$ and sums every primitive orientation b ; the twin-prime endpoint is the single orientation $b = 1$.

1.2 The complete dual mask under a short-shift mean

In the short-shift part we impose the geometric condition that all translated targets occurring on (2) remain in a fixed positive dyadic multiple of X . For example, it is enough that

$$|h_0| + H \max_{v \in \text{supp } G} |v| \leq \frac{\alpha X}{2}. \quad (11)$$

The one-sided version with G supported in $(0, \infty)$ allows the same upper range as TPC-12 after changing the fixed dyadic constants.

For an arbitrary shift-independent mask

$$|\omega_{X,D}(a, b, k)| \leq 1, \quad (12)$$

define

$$\mathcal{Z}_{h,D}^\omega(X) = \sum_{\substack{a,b,k \geq 1, \\ ak > D, \\ (a,b)=1}} \omega_{X,D}(a, b, k) u(ak) \eta(abk^2 + h) W(abk^2/X). \quad (13)$$

The mask may depend on X, D, H and may be discontinuous in every label, but it may not depend on the sampled shift h .

For fixed $0 < \varepsilon < 13/15$, uniformly in

$$X^{2/15+\varepsilon} \leq H \leq X \quad (14)$$

subject to (11), we prove that, for every $A > 0$,

$$\sup_{\|\omega\|_\infty \leq 1} \left| \frac{1}{H} \sum_{q \in \mathbb{Z}} G(q/H) \mathcal{Z}_{h_0+q,D}^\omega(X) \right| \ll_{A,\varepsilon,W,G} X (\log X)^{-A}. \quad (15)$$

This is stronger than uniformity in the two character phases $(a/b)^{it} k^{is}$. It closes that phase family to the entire bounded ℓ^∞ coefficient-mask ball. In particular it includes the exact endpoint mask $\mathbf{1}_{b=k=1}$, a single radial layer $\mathbf{1}_{k=k_0}$, one primitive ray, a dyadic source layer, and arbitrary signed unions of these sets.

The analytic input is Corollary 1.4 of Guth and Maynard [1], which gives the prime number theorem in almost all intervals of length $X^{2/15+\varepsilon}$. Our contribution is an ℓ^2 dual transfer. The translated residual vector has arbitrarily small logarithmic norm, while every bounded GCD–Mellin mask has a divisor-controlled ℓ^2 norm. This is not a new short-interval theorem.

1.3 Endpoint, hard packets, and the information gate

The endpoint mask yields

$$\frac{1}{H} \sum_q G(q/H) \sum_n u(n) \eta(n + h_0 + q) W(n/X) \ll_A X (\log X)^{-A}. \quad (16)$$

After adding back the one-prime terms, its raw form has main term $X I_0(W) I_0(G)$. This reaches the endpoint before taking the mean, but the mean still samples $\asymp H$ shifts. It gives no information about the coordinate $h = 2$.

There is also a legal target-side Vaughan formulation. One first uses the factorized target $t = \ell k$ as the outer center; only then may one apply an arbitrary mask $\omega_{\ell,k}$. The coefficient is independent of the subsequent translation, so (15)'s abstract ℓ^2 mechanism gives a short-shift averaged model for every masked hard packet. Masking a factorization of $r + h$ while retaining r as center would instead make the mask h -dependent and is not covered.

Finally we give two exact boundary theorems. First, nonlinear recovery from a specified local pushforward has minimax radius equal to an ℓ^∞ row-space distance. Second, the complete divisor signature up to R distinguishes primes from composites in $(X, 2X]$ once $R \geq \sqrt{2X}$, whereas below $(1 - \varepsilon)\sqrt{2X}$ a prime and a squarefree semiprime have identical zero signatures. A balanced bilinear factor row separates this null pair exactly. That is a test that the row contains new information, not a proof that its arithmetic sum is small.

1.4 Claim ledger and relation to earlier layers

Result	Label retained	Quantifier	Missing conclusion
TPC-7	complete large-divisor defect	almost all h , $H \geq X^{8/33+\varepsilon}$	prescribed h and individual layers
TPC-13	every (a, b, k) through character phases	signed first moment and squared energy	arbitrary bounded masks
Fixed shell here	exact $k = 1$ shell	every fixed $h \neq 0$	one orientation $b = 1$
Mask theorem here	arbitrary (a, b, k) mask	signed mean, $H \geq X^{2/15+\varepsilon}$	any selected shift

The constructed GCD–Mellin and translation actions below are coordinate representations of finite coefficient arrays. We claim no intrinsic prime, logistic, transfer-operator, or zeta-zero spectrum. We also prove no fixed-shift prime-pair asymptotic, twin-prime lower bound, new level of distribution, or violation of the classical sieve parity barrier.

2 Exact signed selectors in the joint label space

We record the bilinear form of the tensor-Fejér identity from TPC-13. It is the precise sense in which a signed label is retained before any shift average.

Let J be a finite index set with pairwise distinct joint frequencies

$$\xi_j = (\lambda_j, \rho_j) \in \mathbb{R}^2, \quad j \in J, \quad (17)$$

and define

$$\mathcal{O}_x(t, s) = \sum_{j \in J} x_j e^{i(t\lambda_j + s\rho_j)}. \quad (18)$$

Use the normalized Fejér kernel

$$\kappa(t) = \frac{2}{\pi} \left(\frac{\sin(t/2)}{t} \right)^2, \quad \widehat{\kappa}(\xi) = (1 - |\xi|)_+. \quad (19)$$

Proposition 2.1 (Exact dual-mask identity). *Let $T, S > 0$ and suppose that, for every $i \neq j$,*

$$T|\lambda_i - \lambda_j| \geq 1 \quad \text{or} \quad S|\rho_i - \rho_j| \geq 1. \quad (20)$$

For arbitrary coefficient and mask vectors $x, \omega \in \mathbb{C}^J$,

$$\frac{1}{TS} \iint_{\mathbb{R}^2} \kappa(t/T) \kappa(s/S) \mathcal{O}_x(t, s) \overline{\mathcal{O}_\omega(t, s)} dt ds = \sum_{j \in J} x_j \overline{\omega_j}. \quad (21)$$

Conversely, if (20) fails for one pair, (21) fails for some x, ω supported on that pair.

Proof. Expanding the integral, the (i, j) term is multiplied by

$$\widehat{\kappa}(T(\lambda_i - \lambda_j)) \widehat{\kappa}(S(\rho_i - \rho_j)).$$

The diagonal multiplier is 1, and every off-diagonal multiplier vanishes under (20). If that condition fails for one pair, both multipliers are strictly positive; choosing x and ω on the pair leaves a nonzero cross term. $\square$

For the factor coordinates (4), use

$$\lambda_{a,b,k} = \log(a/b), \quad \rho_{a,b,k} = \log k. \quad (22)$$

The global gap bounds in TPC-13 permit $T \asymp X$ and $S \asymp \sqrt{X}$. Sectorwise filters use the smaller anisotropic scales $T_R \asymp R$ and $S_R \asymp \sqrt{X/R}$ after each sector is isolated. We import those gap estimates; no new large-sieve theorem is asserted here.

For a fixed shift h , set

$$x_{a,b,k}(h; X, D) = \mathbf{1}_{ak > D} u(ak) \eta(ak^2 + h) W(ak^2/X). \quad (23)$$

For an arithmetic mask ω , apply (21) with dual vector $\overline{\omega}$. It recovers

$$\sum_{a,b,k \geq 1} x_{a,b,k}(h; X, D) \omega_{a,b,k} \quad (24)$$

without first replacing x by $|x|^2$. In particular,

$$\omega_{a,b,k}^{\text{end}} = \mathbf{1}_{b=k=1}, \quad (25)$$

$$\omega_{a,b,k}^{(k_0)} = \mathbf{1}_{k=k_0}, \quad (26)$$

$$\omega_{a,b,k}^{(a_0, b_0)} = \mathbf{1}_{(a,b)=(a_0, b_0)} \quad (27)$$

select the endpoint, a radial layer, and a primitive ray, respectively.

Proposition 2.2 (Commutation with translation averaging). *Let the mask be independent of h . Under (20),*

$$\begin{aligned} & \frac{1}{H} \sum_q G(q/H) \sum_j x_j(h_0 + q) \overline{\omega_j} \\ &= \frac{1}{HTS} \sum_q G(q/H) \iint_{\mathbb{R}^2} \kappa(t/T) \kappa(s/S) \mathcal{O}_{x(h_0+q)}(t, s) \overline{\mathcal{O}_\omega(t, s)} dt ds. \end{aligned} \quad (28)$$

Proof. All sums are finite and the Fejér kernels are integrable, so the assertion is [proposition 2.1](#) followed by Fubini's theorem. $\square$

The proposition is an exact commuting-filter statement. It does not make a shift-dependent target mask legal. If ω is allowed to inspect h and then retain only factorisations of $r + h$ favorable to primality, the coefficient vector itself changes with the translated von Mangoldt function and the short-interval transfer below does not apply.

The mask symbol $\omega_{a,b,k}$ is unrelated to the standard arithmetic function $\omega(r)$ counting distinct prime divisors. The context will always distinguish them.

3 A fixed-shift law on the exact coprime shell

We now prove (8). The arithmetic function

$$d^*(r) = 2^{\omega(r)} = \#\{(a, b) \in \mathbb{N}^2 : ab = r, (a, b) = 1\} \quad (29)$$

counts unitary divisors. We use $a \parallel r$ to mean $a \mid r$ and $(a, r/a) = 1$.

Lemma 3.1 (Exact shell coefficient). *For every $r \geq 1$,*

$$\sum_{\substack{a \parallel r \\ a > 1}} u(a) = \log \text{rad}(r) - d^*(r) + 1. \quad (30)$$

Consequently

$$\mathcal{K}_h(X) = \sum_{r \geq 1} (\log \text{rad}(r) - d^*(r) + 1) \eta(r+h) W(r/X). \quad (31)$$

Proof. The unitary divisors of $r = \prod_p p^{\nu_p}$ are obtained by choosing, for each $p \mid r$, either 1 or the complete prime-power block p^{ν_p} . Hence there are $2^{\omega(r)}$ of them. The von Mangoldt weight is nonzero on a unitary divisor precisely when that divisor is one prime-power block, and therefore

$$\sum_{a \parallel r} \Lambda(a) = \sum_{p \mid r} \log p = \log \text{rad}(r).$$

Since $u(1) = -1$, deleting $a = 1$ adds 1 and proves (30). Grouping (6) by $r = ab$ proves (31). $\square$

The leading term comes entirely from the mismatch between the ordinary mean of d^* and its mean at a shifted prime.

Theorem 3.2 (Unitary-divisor Titchmarsh input). *For every fixed nonzero integer h ,*

$$\sum_{p \leq y} d^*(p-h) = \beta_h y + O_h\left(\frac{y}{\log y}\right), \quad (32)$$

where $d^*(n) = 0$ for $n \leq 0$ and β_h is given by (7).

This is the $k = 2$ case of Corollary 1.2 of Wang [9]. In the notation of that paper, the displayed constant initially contains

$$\frac{\zeta(2)\zeta(3)}{\zeta(6)} \prod_p \left(1 - \frac{1}{p^2 - p + 1}\right).$$

The local identity

$$\frac{1 - p^{-6}}{(1 - p^{-2})(1 - p^{-3})} \left(1 - \frac{1}{p^2 - p + 1}\right) = 1$$

reduces it to (7).

Lemma 3.3 (The two unitary-divisor means). *For fixed $h \neq 0$,*

$$\sum_{r \geq 1} d^*(r) W(r/X) = \frac{X I_0(W)}{\zeta(2)} \log X + O_W(X), \quad (33)$$

$$\sum_{r \geq 1} d^*(r) \Lambda(r+h) W(r/X) = \beta_h X I_0(W) \log X + O_{h,W}(X). \quad (34)$$

Proof. The Dirichlet series identity

$$\sum_{r \geq 1} \frac{d^*(r)}{r^s} = \frac{\zeta(s)^2}{\zeta(2s)} \quad (\Re s > 1) \quad (35)$$

has a double pole at $s = 1$ with leading coefficient $1/\zeta(2)$. Standard Selberg–Delange summation, or the convolution identity obtained from (35), gives

$$\sum_{r \leq y} d^*(r) = \frac{y}{\zeta(2)} \log y + O(y).$$

Partial summation against $W(r/X)$ proves (33). Only its $X \log X$ coefficient is used; all secondary X terms are absorbed in the remainder.

For (34), first restrict $r + h$ to primes and put $p = r + h$. Partial summation in (32), with the test function

$$y \mapsto \log y W((y - h)/X),$$

gives

$$\sum_p d^*(p - h) \log p W((p - h)/X) = \beta_h X I_0(W) \log X + O_{h,W}(X).$$

Here the main integral is

$$\int_0^\infty \log y W((y - h)/X) dy = X I_0(W) \log X + O_{h,W}(X).$$

The targets $r + h = p^j$, $j \geq 2$, contribute $O_{h,W}(X^{3/4})$, say: there are $O(X^{1/2})$ such targets, and $d^*(p^j - h) \ll_\delta X^\delta$. Taking $\delta = 1/8$ and absorbing the remaining logarithm proves the displayed bound and hence the full von Mangoldt form. $\square$

It remains to show that the logarithmic-radical part has no $X \log X$ term.

Lemma 3.4 (Radical residual). *For every fixed nonzero h ,*

$$\sum_{r \geq 1} \log \text{rad}(r) \eta(r + h) W(r/X) = O_{h,W}(X). \quad (36)$$

Also

$$\sum_{r \geq 1} \eta(r + h) W(r/X) = O_{h,W}(X/\log X). \quad (37)$$

Proof. Write

$$L(r) = \log r - \log \text{rad}(r) = \sum_{\substack{p^j | r \\ j \geq 2}} \log p. \quad (38)$$

The classical zero-free-region prime number theorem, after translating the variable by fixed h and applying smooth partial summation, gives

$$\sum_r \log r \eta(r + h) W(r/X) = O_{h,W}(X). \quad (39)$$

The weaker error shown here is more than sufficient.

We claim

$$\sum_{r \asymp_W X} L(r) (1 + \Lambda(r + h)) \ll_{h,W} X. \quad (40)$$

For the unweighted part, insert (38); the convergent sum $\sum_{p,j \geq 2} (\log p)/p^j$ gives $O(X)$, and the endpoint cost is $O(X^{1/2} \log X)$.

For the von Mangoldt part, first remove proper prime-power targets. Since $L(r) \ll_W \log X$ on the support,

$$\sum_{\substack{\ell^\nu \asymp_W X \\ \nu \geq 2}} (\log \ell) L(\ell^\nu - h) \ll_{h,W} X^{1/2} \log X.$$

It remains to consider prime targets. First take $p^j \leq X^{1/2}$. If $p \nmid h$, the Brun–Titchmarsh inequality in the reduced class $h \pmod{p^j}$ gives [4]

$$\sum_{\substack{r \asymp_W X \\ p^j | r}} \mathbf{1}_{r+h \text{ prime}} \Lambda(r+h) \ll_{h,W} \frac{X}{\varphi(p^j)}.$$

Indeed,

$$\sum_p \sum_{j \geq 2} \frac{\log p}{\varphi(p^j)} = \sum_p \frac{\log p}{(p-1)^2} < \infty.$$

If $p \mid h$, a prime target divisible by p would have to equal p and is absent from the support for large X . For $p^j > X^{1/2}$, use $\Lambda \leq \log(3X)$ and count the $O(X/p^j + 1)$ multiples. Explicitly,

$$\log X \sum_{\substack{j \geq 2, p^j > X^{1/2} \\ p^j \ll_W X}} (\log p) \left(\frac{X}{p^j} + 1 \right) \ll_W X^{3/4} \log X;$$

the squares dominate, while $j \geq 3$ is smaller. This proves (40). Combining it with (39) proves (36).

Finally, (37) is the ordinary smooth prime number theorem, again after translating by fixed h . $\square$

Theorem 3.5 (Fixed-shift coprime-shell law). *Let $h \neq 0$ be fixed. Then*

$$\mathcal{K}_h(X) = \left(\frac{1}{\zeta(2)} - \beta_h \right) X I_0(W) \log X + O_{h,W}(X). \quad (41)$$

For $h = 2$, the leading coefficient is $6/\pi^2 - 1/3$.

Proof. Insert (31). The contribution of $-d^*(r)\eta(r+h)$ is the difference of (33) and (34). The radical and constant terms are $O(X)$ by lemma 3.4. $\square$

Remark 3.6 (Where the fixed shift was bought). The theorem does not estimate each orientation b separately. The exact sum over all b produces $d^*(r)$, and the unitary-divisor Titchmarsh theorem evaluates that aggregate at shifted primes. If one retains only $b = 1$, the sum becomes

$$\sum_a (\Lambda(a) - 1)(\Lambda(a+h) - 1) W(a/X),$$

which is the fixed-shift prime-pair covariance. Thus theorem 3.5 is genuine fixed- h progress at the radial-shell level, not an endpoint estimate in disguise.

4 An ℓ^2 short-shift transfer for the full dual ball

We now change the shift quantifier. The underlying prime theorem is the almost-all short-interval result of Guth and Maynard [1]; the point here is that its exceptional set yields a small vector in the full dual coefficient space.

For integer h_0 and parameters satisfying (14) and (11), define, on the product-center interval in (2),

$$\Delta_{X,H,h_0}(r) = \frac{1}{H} \sum_{q \in \mathbb{Z}} G(q/H) \eta(r + h_0 + q). \quad (42)$$

Theorem 4.1 (Translated prime residual in ℓ^2). *Fix $0 < \varepsilon < 13/15$. For every $A > 0$, uniformly in the stated geometric range,*

$$\left(\sum_{\alpha X \leq r \leq \beta X} |\Delta_{X,H,h_0}(r)|^2 \right)^{1/2} \ll_{A,\varepsilon,W,G} X^{1/2} (\log X)^{-A}. \quad (43)$$

Proof. We give the smoothing argument because G may have two-sided support. First suppose $H \leq X^{0.98}$ and put

$$L = \left\lceil \exp\{(\log X)^{1/8}\} \right\rceil.$$

Partition a fixed interval containing $\text{supp } G$ into L equal pieces. The corresponding integer shift blocks have length $H/L + O(1) \geq X^{2/15+\varepsilon/2}$ for large X , and at most $O(X^{0.98})$. Apply Corollary 1.4 of Guth and Maynard [1] to each forward block. Negative shifts merely change the starting point, which remains comparable with X by (11). Taking the union over $O(L)$ blocks gives a set $\mathfrak{E} \subset [\alpha X, \beta X] \cap \mathbb{Z}$ with

$$|\mathfrak{E}| \ll_{\varepsilon,W,G} X \exp\{-c_{\varepsilon,G}(\log X)^{1/4}\}. \quad (44)$$

Outside this set, the prime number theorem holds on every block with a stretched-exponential relative error. Freeze G on each block. Its C^1 variation contributes $O_G(H \log X/L)$ by the elementary bound $\Lambda \leq \log(3X)$, which is $O_{B,G}(H(\log X)^{-B})$ for every fixed B . Partial summation and the negligible proper prime powers therefore give, outside $\mathfrak{E}$,

$$|\Delta_{X,H,h_0}(r)| \ll_{B,\varepsilon,G} (\log X)^{-B} \quad (45)$$

for every $B > 0$.

If $H > X^{0.98}$, partition the shift range into consecutive blocks of length between $X^{0.97}$ and $2X^{0.97}$. Corollary 1.3 of Guth and Maynard [1], whose all-interval range begins below this length, applies to every block. Freezing G now costs $O_G(X^{0.97} \log X) = O_{B,G}(H(\log X)^{-B})$. Thus (45) holds at every center in the long-window range, and we take $\mathfrak{E} = \emptyset$. This also explains why the claimed range extends beyond the $X^{0.99}$ upper endpoint of the almost-all corollary.

At an exceptional center, the elementary upper-bound sieve, or simply the pointwise bound for Λ , gives

$$|\Delta_{X,H,h_0}(r)| \ll_G \log(3X). \quad (46)$$

Choose $B > A+2$. Squaring (45) and summing over $O(X)$ regular centers gives $O(X(\log X)^{-2A})$. Equations (44)–(46) give a smaller bound on the exceptional set, because the stretched exponential absorbs every fixed logarithmic power. Taking square roots proves the theorem. The $O_G(H^{-1})$ Riemann-sum error in the constant part of $\eta = \Lambda - 1$ is absorbed since H is a fixed positive power of X . $\square$

Corollary 4.2 (Abstract coefficient-space transfer). *Suppose $c_X(r)$ is supported on $[\alpha X, \beta X]$ and, for some fixed $B \geq 0$,*

$$\sum_r |c_X(r)|^2 \ll X(\log X)^B. \quad (47)$$

Assume that the implied constant is independent of X, H, h_0 . Then, for every $A > 0$,

$$\frac{1}{H} \sum_q G(q/H) \sum_r c_X(r) \eta(r + h_0 + q) \ll_{A,B,\varepsilon,W,G,c} X(\log X)^{-A}. \quad (48)$$

Proof. Cauchy–Schwarz and theorem 4.1, requested with $A + B/2 + 2$ in place of A , give the result. $\square$

We now verify that every bounded full-coordinate mask belongs to the same coefficient ball. Group (13) by $r = abk^2$:

$$\mathcal{Z}_{h,D}^\omega(X) = \sum_r C_{\omega,D}(r) \eta(r+h), \quad (49)$$

where

$$C_{\omega,D}(r) = W(r/X) \sum_{\substack{abk^2=r, (a,b)=1 \\ ak>D}} \omega_{X,D}(a,b,k) u(ak). \quad (50)$$

Lemma 4.3 (Uniform bounded-mask ledger). *Uniformly in $D \geq 1$ and every mask satisfying (12),*

$$|C_{\omega,D}(r)| \leq \|W\|_\infty (\log r + \tau(r)), \quad (51)$$

$$\sum_r |C_{\omega,D}(r)|^2 \ll_W X (\log X)^3. \quad (52)$$

Proof. The bijection (4) sends the triples with $abk^2 = r$ to the factor pairs (n, m) with $nm = r$, and $ak = n$. Therefore

$$|C_{\omega,D}(r)| \leq \|W\|_\infty \sum_{n|r} |\Lambda(n) - 1| \leq \|W\|_\infty \left(\sum_{n|r} \Lambda(n) + \tau(r) \right).$$

The divisor identity $\sum_{n|r} \Lambda(n) = \log r$ proves (51). The classical second moment [6]

$$\sum_{r \leq y} \tau(r)^2 \ll y (\log(2y))^3$$

then proves (52). $\square$

Theorem 4.4 (Uniform full-label mask transfer). *Fix $0 < \varepsilon < 13/15$. For every $A > 0$, uniformly in $D \geq 1$, every mask (12), and the range of theorem 4.1,*

$$\frac{1}{H} \sum_q G(q/H) \mathcal{Z}_{h_0+q,D}^\omega(X) \ll_{A,\varepsilon,W,G} X (\log X)^{-A}. \quad (53)$$

The implied constant is independent of the mask.

Proof. Apply corollary 4.2 to (50), using (52). $\square$

There is an equivalent raw-target formulation. Put

$$\mathcal{T}_{h,D}^\omega(X) = \sum_{\substack{a,b,k \geq 1, (a,b)=1 \\ ak>D}} \omega_{X,D}(a,b,k) u(ak) \Lambda(abk^2+h) W(abk^2/X), \quad (54)$$

$$\mathcal{M}_D^\omega(X) = \sum_{\substack{a,b,k \geq 1, (a,b)=1 \\ ak>D}} \omega_{X,D}(a,b,k) u(ak) W(abk^2/X). \quad (55)$$

Corollary 4.5 (Uniform raw mask model). *Under the hypotheses of theorem 4.4,*

$$\frac{1}{H} \sum_q G(q/H) \mathcal{T}_{h_0+q,D}^\omega(X) = I_0(G) \mathcal{M}_D^\omega(X) + O_{A,\varepsilon,W,G}(X (\log X)^{-A}). \quad (56)$$

Proof. Write $\Lambda = 1 + \eta$ and use theorem 4.4. Euler summation gives $H^{-1} \sum_q G(q/H) = I_0(G) + O_G(H^{-1})$. The pointwise ledger and the mean order of τ give $|\mathcal{M}_D^\omega| \ll_W X \log(2X)$, so the Riemann-sum error is absorbed in the displayed remainder. $\square$

Remark 4.6 (Uniform masks are not uniform target adaptation). The theorem permits a different mask for every X, D, H , including masks chosen after inspecting the source labels. It does not permit $\omega = \omega_h$ to inspect each translated target. Such a family could retain precisely the prime targets and would make the assertion circular.

5 Endpoint consequences and parity-conditioned shifts

The endpoint is selected before the shift average by (25). If $D \leq X^{1-\eta_0}$ for a fixed $\eta_0 > 0$, then $a > D$ automatically on the support of $W(a/X)$ for large X , and

$$\mathcal{Z}_h^{\text{end}}(X) = \sum_{n \geq 1} u(n) \eta(n+h) W(n/X). \quad (57)$$

Corollary 5.1 (Signed endpoint mean at the two-fifteenths scale). *Under the hypotheses of theorem 4.4, for every $A > 0$,*

$$\frac{1}{H} \sum_q G(q/H) \sum_n u(n) \eta(n+h_0+q) W(n/X) \ll_{A,\varepsilon,W,G} X (\log X)^{-A}. \quad (58)$$

Proof. Take $D = 1$ and $\omega = \omega^{\text{end}}$ in theorem 4.4. □

The raw form is a smooth average of prime-pair correlations.

Corollary 5.2 (Raw endpoint prime-pair mean). *Under the same hypotheses,*

$$\begin{aligned} \frac{1}{H} \sum_q G(q/H) \sum_n \Lambda(n) \Lambda(n+h_0+q) W(n/X) \\ = X I_0(W) I_0(G) + O_{A,\varepsilon,W,G}(X (\log X)^{-A}). \end{aligned} \quad (59)$$

Proof. Apply corollary 4.2 to $c_X(n) = \Lambda(n) W(n/X)$. Its hypothesis follows from the classical estimate

$$\sum_n \Lambda(n)^2 |W(n/X)|^2 \ll_W X \log X.$$

This controls the centered target. Now use

$$\frac{1}{H} \sum_q G(q/H) = I_0(G) + O_G(H^{-1})$$

and the smooth prime number theorem $\sum_n \Lambda(n) W(n/X) = X I_0(W) + O_A(X (\log X)^{-A})$. The polynomial lower bound on H absorbs the Riemann-sum error. □

This corollary has a different quantifier from the almost-all-shift theorem of Matomäki et al. [5]. That theorem gives the singular-series asymptotic at almost every individual shift in its long window once $H \geq X^{8/33+\varepsilon}$. On the overlapping unmasked range it is stronger and yields such a mean. Equation (59) reaches the shorter two-fifteenths scale and the ambient theorem permits arbitrary source masks, but it gives no individual-shift asymptotic.

5.1 Even shifts and the correct local center

Conditioning the shift on parity changes the local mean. Suppose h_0 is even and sum only over even q . For a product center r , define

$$\eta_r^{(2)}(h) = \Lambda(r+h) - 2\mathbf{1}_{2 \nmid r}, \quad h \equiv 0 \pmod{2}. \quad (60)$$

If r is odd, the target runs over odd integers and has conditional von Mangoldt density 2; if r is even, the target is even and only powers of 2 contribute.

$$C_H = \frac{1}{H} \sum_q G(q/H), \quad C_H^{(2)} = \frac{2}{H} \sum_{q \equiv 0 \pmod{2}} G(q/H), \quad (61)$$

$$\Delta^{(2)}(r) = \frac{2}{H} \sum_{q \equiv 0 \pmod{2}} G(q/H) \eta_r^{(2)}(h_0+q). \quad (62)$$

Theorem 5.3 (Uniform even-shift mask transfer). *Under the hypotheses of [theorem 4.4](#), with h_0 even, for every $A > 0$,*

$$\sup_{\|\omega\|_\infty \leq 1} \left| \frac{2}{H} \sum_{\substack{q \in \mathbb{Z} \\ q \equiv 0 \pmod{2}}} G(q/H) \sum_{\substack{a, b, k \geq 1, \\ (a, b) = 1 \\ ak > D}} \omega_{X, D}(a, b, k) u(ak) \eta_{abk^2}^{(2)}(h_0 + q) W(abk^2/X) \right| \ll_{A, \varepsilon, W, G} X(\log X)^{-A}. \quad (63)$$

Proof. Euler summation gives

$$C_H = I_0(G) + O_G(H^{-1}), \quad C_H^{(2)} = I_0(G) + O_G(H^{-1}).$$

If r is odd, then odd q make $r + h_0 + q$ even. At height comparable with X , the von Mangoldt support on that lattice contains only $O_G(1)$ powers of 2. Consequently

$$\Delta^{(2)}(r) = 2\Delta_{X, H, h_0}(r) + O_G\left(\frac{\log X}{H}\right).$$

If r is even, the even- q target itself is even and

$$\Delta^{(2)}(r) = O_G\left(\frac{\log X}{H}\right).$$

The polynomial lower bound for H therefore gives $\|\Delta^{(2)}\|_2 \ll_A X^{1/2}(\log X)^{-A}$. Apply the mask ledger ([52](#)) and Cauchy–Schwarz. $\square$

Corollary 5.4 (Conditional even prime-pair mean). *Under the hypotheses of [theorem 5.3](#), one has*

$$\begin{aligned} \frac{2}{H} \sum_{\substack{q \in \mathbb{Z} \\ q \equiv 0 \pmod{2}}} G(q/H) \sum_n \Lambda(n) \Lambda(n + h_0 + q) W(n/X) \\ = 2XI_0(W)I_0(G) + O_{A, \varepsilon, W, G}(X(\log X)^{-A}). \end{aligned} \quad (64)$$

Proof. Let

$$P^{(2)}(n) = \frac{2}{H} \sum_{q \equiv 0 \pmod{2}} G(q/H) \Lambda(n + h_0 + q).$$

The preceding proof gives $P^{(2)}(n) = \Delta^{(2)}(n) + 2C_H^{(2)} \mathbf{1}_{2 \nmid n}$. By Cauchy–Schwarz, the contribution of $\Delta^{(2)}$ against $\Lambda(n)W(n/X)$ is $O_A(X(\log X)^{-A})$ after requesting a larger value of A . The smooth prime number theorem, with the powers of 2 removed, gives

$$\sum_{2 \nmid n} \Lambda(n) W(n/X) = XI_0(W) + O_A(X(\log X)^{-A}).$$

Since $C_H^{(2)} = I_0(G) + O_G(H^{-1})$, the asserted main term follows. $\square$

The factor 2 is the conditional mean of the prime-pair singular series on even shifts. Without the normalization $2/H$, the leading coefficient is 1, the same as in ([59](#)). Neither normalization selects the particular even shift 2.

6 Masked Vaughan packets after target recentering

The full-label theorem controls factorisations of the source center $r = mn$. A target-side Vaughan packet requires a different ordering of operations: first make the factorized target itself the center, and only then average its translated prime partner.

For a finitely supported function F and parameters $U, V \geq 1$, put

$$\beta_V(k) = \sum_{\substack{d|k \\ d \leq V}} \mu(d). \quad (65)$$

The truncated Möbius coefficient $\beta_V(k)$ is unrelated to the fixed-shift Euler factor β_h in (7). Vaughan's identity in the form used in TPC-6 [12] is

$$\begin{aligned} \sum_t \Lambda(t)F(t) &= \sum_{t \leq U} \Lambda(t)F(t) - \sum_{\substack{\ell \leq U \\ d \leq V}} \Lambda(\ell)\mu(d) \sum_{j \geq 1} F(\ell dj) \\ &\quad + \sum_{d \leq V} \mu(d) \sum_{j \geq 1} (\log j)F(dj) - \sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell)\beta_V(k)F(\ell k). \end{aligned} \quad (66)$$

Moreover,

$$\beta_V(k) = 0 \quad (1 < k \leq V), \quad (67)$$

so the last sum is supported on $\ell > U, k > V$. The identity itself is classical [3, 8]; we include it to fix signs and the factor labels.

Consider the target-weighted prime-pair form

$$\mathcal{C}_h^{\text{tar}}(X) = \sum_t \Lambda(t)\Lambda(t+h)W(t/X). \quad (68)$$

Taking $F(t) = \Lambda(t+h)W(t/X)$ in (66), the last term is the hard packet. We permit an arbitrary bounded mask on its now fixed factorisation.

Definition 6.1 (Masked hard-packet coefficient). Let $|\omega_{\ell,k}| \leq 1$ be independent of h . Define

$$c_{U,V}^\omega(t) = -W(t/X) \sum_{\substack{\ell k = t \\ \ell > U, k > V}} \omega_{\ell,k} \Lambda(\ell) \beta_V(k), \quad (69)$$

$$\mathcal{H}_{h;U,V}^\omega(X) = \sum_t c_{U,V}^\omega(t) \Lambda(t+h), \quad (70)$$

$$\mathcal{H}_{h;U,V}^{\omega,\circ}(X) = \sum_t c_{U,V}^\omega(t) \eta(t+h). \quad (71)$$

The unmasked case $\omega \equiv 1$ is exactly the final term of Vaughan's identity for (68). A mask extracts a specified dyadic or signed subpacket; the extraction is made before the shift is sampled.

Lemma 6.2 (Uniform hard-packet ledger). *Uniformly in $U, V \geq 1$ and $\|\omega\|_\infty \leq 1$,*

$$|c_{U,V}^\omega(t)| \leq \|W\|_\infty \tau(t) \log t, \quad (72)$$

$$\sum_t |c_{U,V}^\omega(t)|^2 \ll_W X (\log X)^5. \quad (73)$$

Proof. Equation (65) gives $|\beta_V(k)| \leq \tau(k) \leq \tau(t)$ when $k | t$. Hence

$$|c_{U,V}^\omega(t)| \leq \|W\|_\infty \tau(t) \sum_{\ell|t} \Lambda(\ell) = \|W\|_\infty \tau(t) \log t.$$

The second assertion follows from the classical second moment of τ . $\square$

Theorem 6.3 (Short-shift averaged closure of every masked hard packet). *Fix $0 < \varepsilon < 13/15$. Under the range of [theorem 4.1](#), uniformly in U, V and every bounded shift-independent mask,*

$$\frac{1}{H} \sum_q G(q/H) \mathcal{H}_{h_0+q;U,V}^{\omega,\circ}(X) \ll_{A,\varepsilon,W,G} X(\log X)^{-A} \quad (74)$$

for every $A > 0$. Equivalently,

$$\begin{aligned} \frac{1}{H} \sum_q G(q/H) \mathcal{H}_{h_0+q;U,V}^{\omega}(X) &= I_0(G) \sum_t c_{U,V}^{\omega}(t) \\ &\quad + O_{A,\varepsilon,W,G}(X(\log X)^{-A}). \end{aligned} \quad (75)$$

Proof. Apply [corollary 4.2](#) with $c_X = c_{U,V}^{\omega}$ and use [\(73\)](#). The raw form follows from $\Lambda = 1 + \eta$ and Euler summation. Indeed, $|\sum_t c_{U,V}^{\omega}(t)| \ll_W X(\log X)^{5/2}$ by Cauchy–Schwarz, so the $O_G(H^{-1})$ lattice error is absorbed by the polynomial lower bound on H . $\square$

Remark 6.4 (What has and has not become Type II). The coefficient in [\(69\)](#) is the bilinear hard packet left by Vaughan’s identity, and the theorem controls every one of its fixed masks. A balanced dyadic Type-II packet is obtained only after placing both factors in comparable prescribed ranges. The cancellation in [\(74\)](#), however, comes from averaging the second prime variable and importing the Guth–Maynard short-interval theorem. We do not prove a new fixed- h Type-II estimate.

Remark 6.5 (Why target recentering is mandatory). If r remains the outer center and one masks a factorisation of $r + h$, the mask varies with h and can align with the target primes. Such a packet is not covered by [theorem 4.1](#). In [\(69\)](#), $t = \ell k$ is fixed first; only the partner $t + h$ is translated. This is the legal order of operations.

7 The local-information null space and the Type-II gate

The preceding theorems preserve arbitrary labels that are already present in a coefficient array. They do not say whether a proposed label family contains enough information to distinguish primes from balanced semiprimes. We make that question exact.

7.1 An exact minimax formula for a local pushforward

Let I and Σ be finite sets, let $\sigma : I \rightarrow \Sigma$ be a signature map, and let $w = (w_i)_{i \in I}$ be real weights. The associated pushforward is

$$(\mathcal{N}_{\sigma,w}x)(s) = \sum_{\substack{i \in I \\ \sigma(i)=s}} w_i x_i, \quad s \in \Sigma. \quad (76)$$

For $v \in \mathbb{R}^I$, write $L_v(x) = \sum_i v_i x_i$.

Theorem 7.1 (Exact local-data minimax radius). *For $B > 0$, even allowing an arbitrary nonlinear decoder $\Phi : \text{ran } \mathcal{N}_{\sigma,w} \rightarrow \mathbb{R}$, one has*

$$\begin{aligned} \inf_{\Phi} \sup_{\|x\|_1 \leq B} |L_v(x) - \Phi(\mathcal{N}_{\sigma,w}x)| \\ = B \inf_{g: \Sigma \rightarrow \mathbb{R}} \max_{i \in I} |v_i - w_i g(\sigma(i))|. \end{aligned} \quad (77)$$

Consequently exact recovery on the ℓ^1 ball is possible if and only if

$$v_i = w_i g(\sigma(i)) \quad (i \in I) \quad (78)$$

for some g . In the unweighted case $w_i = 1$,

$$e_B(v; \sigma) = \frac{B}{2} \max_{s \in \Sigma} \left(\max_{\sigma(i)=s} v_i - \min_{\sigma(i)=s} v_i \right). \quad (79)$$

Empty fibers are omitted from the maximum.

Proof. Let $N = \mathcal{N}_{\sigma, w}$. Two points of the ℓ^1 ball with the same observation differ by an element of $\ker N$ of norm at most $2B$. Conversely, if $z \in \ker N$ and $\|z\|_1 \leq 2B$, then $z/2$ and $-z/2$ are such a pair. The smallest worst-case radius of all observation fibers is therefore

$$B \sup_{\substack{z \in \ker N \\ \|z\|_1 \leq 1}} |L_v(z)|.$$

Finite-dimensional quotient duality identifies this norm with

$$B \operatorname{dist}_{\ell^\infty}(v, \operatorname{ran} N^*).$$

Since $(N^*g)_i = w_i g(\sigma(i))$, this is (77). The zero-distance criterion gives (78). When $w_i = 1$, the best constant on each fiber is the midpoint of the extreme v_i values, proving (79). $\square$

Thus adding nonlinear post-processing does not repair a missing row. The obstruction is an exact null-space statement, not a limitation of a particular learning rule or inversion algorithm.

7.2 The square-root divisor-information dichotomy

For $t \in I_X = (X, 2X] \cap \mathbb{N}$ and $R \geq 2$, define the complete local divisor signature

$$\sigma_R(t) = (\mathbf{1}_{d|t}, d < t)_{2 \leq d \leq R}. \quad (80)$$

Thus only proper divisors are recorded, even if $R \geq t$.

Proposition 7.2 (Square-root information gate). *The following statements hold.*

- (i) If $R \geq \sqrt{2X}$, then $\sigma_R(t) = 0$ for $t \in I_X$ if and only if t is prime.
- (ii) Fix $0 < \delta < 1$. For all sufficiently large X , uniformly for $R \leq (1 - \delta)\sqrt{2X}$, there are a prime $p \in I_X$ and distinct primes $q_1, q_2 > R$ such that

$$s = q_1 q_2 \in I_X, \quad \sigma_R(p) = \sigma_R(s) = 0. \quad (81)$$

Proof. If $t \leq 2X$ is composite, it has a divisor d with $2 \leq d \leq \sqrt{t} \leq \sqrt{2X}$, proving (i).

For (ii), put

$$c_0 = \max\{1, (1 - \delta)\sqrt{2}\} < \sqrt{2}$$

and choose a closed interval $[c_1, c_2]$ strictly inside $(c_0, \sqrt{2})$. The prime number theorem supplies two distinct primes in $[c_1\sqrt{X}, c_2\sqrt{X}]$ for all sufficiently large X ; after shrinking the interval if necessary, every product of two points in it lies in $(X, 2X]$. Call the primes q_1, q_2 . They exceed R , so their product has zero signature. A prime $p \in (X, 2X]$, again supplied by the prime number theorem, has the same signature. $\square$

This is an information-resolution threshold. It is not an analytic level of distribution and does not assert that sieving to $\sqrt{2X}$ can be carried out with a sufficiently accurate remainder. It is a theorem about the specified data interface σ_R : an algorithm supplied with the integer t itself or with additional arithmetic rows has more information. Its relation to the classical parity limitation is therefore diagnostic, not an alternative proof of that limitation [7].

Take the unweighted pushforward by σ_R and view an unknown $x \in \ell^1(I_X)$ as a signed mass distribution. On the null pair in (81), the prime indicator takes values 1 and 0, the Liouville function takes values -1 and 1 , and Λ takes values $\log p$ and 0 . Hence [theorem 7.1](#) gives, on the unit ball,

$$e_1(\mathbf{1}_{\mathbb{P}}; \sigma_R) = \frac{1}{2}, \quad (82)$$

$$e_1(\lambda; \sigma_R) = 1, \quad (83)$$

$$e_1(\Lambda; \sigma_R) \geq \frac{1}{2} \log p. \quad (84)$$

The first two equalities also use the global ranges $\mathbf{1}_{\mathbb{P}} \in \{0, 1\}$ and $\lambda \in \{-1, 1\}$.

7.3 A balanced factor row sees the null direction

For coefficient sequences α, β , define the bilinear observation on a response vector x by

$$\mathcal{B}_{\alpha, \beta}(x) = \sum_{m, n} \alpha_m \beta_n x(mn). \quad (85)$$

For the pair in (81),

$$\mathcal{B}_{\delta_{q_1}, \delta_{q_2}}(\delta_p) = 0, \quad \mathcal{B}_{\delta_{q_1}, \delta_{q_2}}(\delta_s) = 1. \quad (86)$$

Thus a single balanced factor row strictly enlarges this specified divisor-signature data model: it separates two points that every test $d \leq R$ identifies.

The same distinction is visible in the hard coefficient from (66). Remove the smooth weight and put

$$H_{U, V}(N) = - \sum_{\substack{\ell k = N \\ \ell > U, k > V}} \Lambda(\ell) \beta_V(k). \quad (87)$$

If $p, q_1, q_2 > \max\{U, V\}$ and $s = q_1 q_2$, then

$$H_{U, V}(p) = 0, \quad H_{U, V}(s) = -\log s. \quad (88)$$

Indeed $\beta_V(q_i) = 1$. In Vaughan's identity the $d = 1$ logarithmic packet contributes $+\log N$ to both inputs; the hard packet cancels it exactly for the semiprime and leaves it untouched for the prime.

Equations (86) and (88) certify that this balanced bilinear row contains information absent from the specified local signature. They provide no cancellation estimate for a long Type-II sum. That analytic estimate is the gate that remains.

8 Why the positive theorems do not select a fixed prime pair

There are two different losses in the preceding results. The fixed-shift law retains the shell $k = 1$ but aggregates its orientations. The full-mask law retains every orientation but averages the shift. Neither loss can be reversed by an algebraic change of coordinates alone.

8.1 One shift average does not recover one shift coordinate

The following Hilbert-space calculation isolates the quantifier loss. Let Q be a finite shift set, let $w \in \mathbb{C}^Q \setminus \{0\}$, and observe only $A_w(z) = \langle z, w \rangle = \sum_{q \in Q} z_q \overline{w_q}$.

Proposition 8.1 (Exact coordinate-recovery minimax). *For $q_* \in Q$ and $B > 0$,*

$$\begin{aligned} & \inf_{\Phi: \mathbb{C} \rightarrow \mathbb{C}} \sup_{\|z\|_2 \leq B} |z_{q_*} - \Phi(A_w(z))| \\ &= B \left(1 - \frac{|w_{q_*}|^2}{\|w\|_2^2} \right)^{1/2}. \end{aligned} \quad (89)$$

Proof. As in [theorem 7.1](#), the exact radius is B times the norm of the target functional on $\ker A_w$. This is the distance from the coordinate vector e_{q_*} to $\text{span}\{w\}$, which is the right side of (89). The orthogonal projection also gives the optimal linear decoder, so allowing nonlinear decoders does not reduce the radius. $\square$

For the smooth weights in this paper,

$$w_q = H^{-1}G(q/H), \quad \|w\|_2^2 \asymp_G H^{-1}. \quad (90)$$

If the selected coordinate lies in the support, then $|w_{q_*}|^2 \ll_G H^{-2}$, and therefore the minimax radius is

$$B(1 - O_G(H^{-1})). \quad (91)$$

If the coordinate lies outside the sampled window, then $w_{q_*} = 0$ and the radius is exactly B . On an unrestricted response ball, a growing smooth mean retains asymptotically none of the stable information needed to recover one coordinate. Arithmetic prime-pair vectors are not arbitrary vectors, so this proposition is not an impossibility theorem for twin primes. It says precisely why the averaged theorem, by itself, cannot be inverted at $h = 2$.

8.2 Why the fixed shell does not isolate the endpoint

At fixed h , the $k = 1$ shell has the exact orientation decomposition

$$\mathcal{K}_h(X) = \sum_{\substack{a>1, b\geq 1 \\ (a,b)=1}} u(a)\eta(ab+h)W(ab/X). \quad (92)$$

The proof of [theorem 3.5](#) first sums all orientations with $ab = r$, producing the unitary-divisor coefficient $\log \text{rad}(r) - d^*(r) + 1$. Biao Wang's theorem estimates that aggregate. It supplies no estimate uniform in a dual orientation mask and therefore cannot be combined with the exact Fejér selector to retain only $b = 1$. For that mask the target is exactly

$$\sum_a (\Lambda(a) - 1)(\Lambda(a+h) - 1)W(a/X), \quad (93)$$

the unresolved fixed-shift covariance.

The Hardy–Littlewood prime-pair conjecture predicts the local main-term coefficient

$$\sum_a (\Lambda(a) - 1)(\Lambda(a+h) - 1)W(a/X) \sim (\mathfrak{S}(h) - 1)XI_0(W), \quad (94)$$

where $\mathfrak{S}(h)$ is the prime-pair singular series [2]. In particular, $\mathfrak{S}(2) = 2C_2$, with

$$C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2} \right).$$

Thus the expected $h = 2$ covariance is $(2C_2 - 1)XI_0(W)$, not zero.

This distinction is easy to lose: exact coordinate recovery is an identity for a finite coefficient array, while an arithmetic asymptotic is a norm estimate uniform over the selector needed for inversion. The former does not manufacture the latter.

8.3 Sparse rays fall outside the natural-scale transfer

The ℓ^2 transfer is uniform over masks, but its absolute bound need not be sharp at the natural size of every sparse mask. Fix a primitive orientation (a_0, b_0) . Its product centers have the form

$$r = a_0 b_0 k^2 \asymp X, \quad (95)$$

so only $O_{a_0, b_0}(X^{1/2})$ centers occur. The source coefficient has

$$\|c_{a_0, b_0}\|_2 \ll_{a_0, b_0, W} X^{1/4} \log X. \quad (96)$$

Combining this with the global residual norm (43) yields only

$$\left| \sum_{r=a_0 b_0 k^2} c_{a_0, b_0}(r) \Delta_{X, H, h_0}(r) \right| \ll_A X^{3/4} (\log X)^{-A}, \quad (97)$$

after renaming A . For every fixed A , this is not $o(X^{1/2})$. Thus the uniform theorem controls the full mask ball on the ambient X -scale, but it does not give natural-scale cancellation on an individual quadratic ray. Such a result would require a restriction estimate for the translated residual on sparse polynomial centers.

8.4 The remaining fixed-shift gate

The results can now be placed in a strict logical order:

- (1) the joint GCD–Mellin coordinate preserves every finite factor label exactly;
- (2) the fixed $k = 1$ aggregate is evaluable at every prescribed $h \neq 0$;
- (3) all shift-independent signed labels are controllable after a growing short-shift mean;
- (4) local divisor labels below the square-root gate possess an exact prime/semiprime null direction;
- (5) a balanced bilinear row detects that direction, but estimating the corresponding fixed- h arithmetic packet remains open.

Accordingly, the next useful target is not another algebraic inversion. It is a fixed-shift asymptotic, or cancellation after subtracting the correct local main term, for a sufficiently localized Type-II packet with enough uniformity to preserve the endpoint label. Proving such an estimate would be genuine analytic progress. The present paper neither assumes nor claims it.

9 Finite certificate and conclusion

The accompanying script `experiments/signed_label_certificate.py` is a deterministic finite certificate. It checks the following identities independently of the asymptotic proofs:

- (i) the unitary-divisor shell coefficient $\sum_{a \parallel r, a > 1} (\Lambda(a) - 1) = \log \text{rad}(r) - 2^{\omega(r)} + 1$ on a finite range;
- (ii) the bijection between factor pairs (n, m) and coordinates (a, b, k) , including endpoint and radial masks;
- (iii) Vaughan’s identity and the hard-packet values on the finite prime/semiprime witness;
- (iv) the identical divisor signatures of

$$p = 101, \quad s = 11 \cdot 13 = 143, \quad R = 10, \quad (98)$$

together with their separation by a balanced factor row;

- (v) the exact finite-fiber minimax radius and the Hilbert-space shift-coordinate radius for sample weights.

The script writes a JSON record with all parameters and residuals, and the unit test recomputes every assertion. The certificate does not test the Guth–Maynard or unitary-divisor Titchmarsh inputs.

The two quantifier regimes may be summarized without implication arrows:

	All orientations b	Endpoint orientation $b = 1$
Fixed h	Fixed coprime-shell law in theorem 3.5	Unresolved fixed prime-pair covariance
Growing mean in h	Full bounded-mask transfer	Endpoint corollaries in corollaries 5.1 and 5.2

The main conclusions are therefore separated cleanly by quantifier. At every fixed nonzero shift, one nontrivial exact radial shell has an explicit asymptotic; at $h = 2$ its leading coefficient is $6/\pi^2 - 1/3$. Under a growing short-shift average of length $H \geq X^{2/15+\varepsilon}$, every bounded shift-independent label mask is controlled simultaneously, including the prime-pair endpoint and every masked target-recentered Vaughan packet. An exact minimax theorem then locates the information missing from local divisor signatures, and a balanced bilinear row detects the resulting prime/semiprime null direction.

What remains is analytic rather than representational: obtain a fixed-shift asymptotic, or cancellation after subtracting its correct local main term, for a sharply localized Type-II packet while retaining the endpoint label. The fixed-shell law, the full-mask short-shift theorem, and the null-space witness provide three independently checkable footholds for that next step. None is a fixed-shift prime-pair asymptotic or a breach of the parity barrier.

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